

■ 3.5.6 Indefinite Integrals

The *Mathematica* function `Integrate[f, x]` gives you the *indefinite integral* $\int f \, dx$. You can think of the process of indefinite integration as like an inverse of differentiation. If you take the result from `Integrate[f, x]`, and then differentiate it, you always get a result that is mathematically equal to the original expression f .

In general, however, there is a whole family of results which have the property that their derivative is f . `Integrate[f, x]` gives you an expression whose derivative is f . You can get other expressions by adding an arbitrary constant of integration.

When you fill in explicit limits for your integral, any such constants of integration must cancel out. Even though the indefinite integral contains arbitrary constants, it is still often very convenient to manipulate it before filling in the limits.

Mathematica applies standard rules to find indefinite integrals.

```
In[1]:= Integrate[x^2, x]
```

$$\text{Out}[1]= \frac{x^3}{3}$$

You can add an arbitrary constant to the indefinite integral, and still get the same derivative. `Integrate` simply gives you an expression with the required derivative.

```
In[2]:= D[%, x]
```

$$\text{Out}[2]= x^2$$

This gives the indefinite integral $\int \frac{dx}{x^2-1}$.

```
In[3]:= Integrate[1/(x^2 - 1), x]
```

$$\text{Out}[3]= \frac{\text{Log}[-1 + x]}{2} - \frac{\text{Log}[1 + x]}{2}$$

Differentiating should give the original function back again.

```
In[4]:= D[%, x]
```

$$\text{Out}[4]= \frac{1}{2(-1 + x)} - \frac{1}{2(1 + x)}$$

You need to manipulate it to get it back into the original form.

```
In[5]:= Simplify[%]
```

$$\text{Out}[5]= \frac{1}{-1 + x^2}$$

The `Integrate` function assumes that any object that does not explicitly contain the integration variable is independent of it, and can be treated as a constant. As a result, `Integrate` is like an inverse of the *partial differentiation* function `D`.

The variable a is assumed to be independent of x

```
In[6]:= Integrate[a x^2, x]
Out[6]=  $\frac{a x^3}{3}$ 
```

The integration variable can be any expression.

```
In[7]:= Integrate[x b[x]^2, b[x]]
Out[7]=  $\frac{x b[x]^3}{3}$ 
```

Another assumption that **Integrate** implicitly makes is that all the symbolic quantities in your integrand have “generic” values. Thus, for example, *Mathematica* will tell you that $\int x^n dx$ is $\frac{x^{n+1}}{n+1}$ even though this is not true in the special case $n = -1$.

Mathematica gives the standard result for this integral, implicitly assuming that n is not equal to -1 .

```
In[8]:= Integrate[x^n, x]
Out[8]=  $\frac{x^{1+n}}{1+n}$ 
```

If you specifically give an exponent of -1 , *Mathematica* produces a different result.

```
In[9]:= Integrate[x^-1, x]
Out[9]= Log[x]
```

The results you get by doing integrals can often be written in many different forms. *Mathematica* follows various principles in deciding which form to use in each particular case. One principle is that if your input does not contain complex numbers, then neither should your output. Thus, for example, *Mathematica* tries to choose between forms involving logarithms and arc tangents in such a way as to avoid unnecessary complex numbers.

Mathematica writes this integral in terms of logarithms.

```
In[10]:= Integrate[1/(x^2 - 8), x]
Out[10]=  $\frac{\text{Log}\left[\frac{4 \sqrt{2} - 2 x}{4 \sqrt{2} + 2 x}\right]}{4 \sqrt{2}}$ 
```

If you reverse the sign of the denominator term, *Mathematica* gives a result in terms of arc tangents.

```
In[11]:= Integrate[1/(x^2 + 8), x]
Out[11]=  $\frac{\text{ArcTan}\left[\frac{x}{2 \sqrt{2}}\right]}{2 \sqrt{2}}$ 
```

If your integral involves symbolic parameters with unspecified signs, it is not so clear how to avoid complex numbers. In this case, *Mathematica* gives the result appropriate for positive a .

```
In[12]:= Integrate[1/(x^2 + a), x]
          ArcTan[ $\frac{x}{\sqrt{a}}$ ]
Out[12]=  $\frac{\text{ArcTan}[\frac{x}{\sqrt{a}}]}{\sqrt{a}}$ 
```

In the course of evaluating the integral, *Mathematica* explicitly calls `Sign` to try and determine the sign of a . You can add your own definition for `Sign[a]`.

```
In[13]:= a/: Sign[a] = -1
Out[13]= -1
```

Now *Mathematica* will instead assume that a is negative, and will give the result in the appropriate form.

```
In[14]:= Integrate[1/(x^2 + a), x]
Out[14]= 
$$\frac{-\frac{1}{2} (\text{Log}[-2 \sqrt{a} + 2 x] - \text{Log}[2 \sqrt{a} + 2 x])}{\sqrt{a}}$$

```